Лабораторная работа №12

Синхронизация времени

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Группа: НФИбд 02–23

дисциплина: **Администрирование сетевых подсистем (Lab 12)**

Цель работы

Целью данной работы является получение навыков по управлению системным временем и настройке синхронизации времени.

root@server:~

```
[vagrant@server ~]$ sudo -i
[root@server ~]# timedatectl
               Local time: Thu 2026-03-12 08:34:52 UTC
           Universal time: Thu 2026-03-12 08:34:52 UTC
                 RTC time: Thu 2026-03-12 08:34:52
                Time zone: UTC (UTC, +0000)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
[root@server ~]# date
Thu Mar 12 08:36:20 AM UTC 2026
[root@server ~]# hwclock
2026-03-12 08:36:28.246236+00:00
[root@server ~]#
```

root@server:~

```
2026-03-12 08:36:28.246236+00:00
[root@server ~]# dnf -y install chrony
Rocky Linux 9 - BaseOS                          8.9 kB/s | 4.3 kB     00:00
Rocky Linux 9 - BaseOS                          5.7 MB/s |  16 MB     00:02
Rocky Linux 9 - AppStream                        12 kB/s | 4.8 kB     00:00
Rocky Linux 9 - AppStream                       6.9 MB/s |  17 MB     00:02
Rocky Linux 9 - Extras                          5.9 kB/s | 3.1 kB     00:00
Rocky Linux 9 - Extras                           18 kB/s |  17 kB     00:00
Package chrony-4.5-1.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package          Architecture     Version               Repository        Size
================================================================================
Upgrading:
 chrony           x86_64           4.6.1-2.el9           baseos           340 k

Transaction Summary
================================================================================
Upgrade  1 Package

Total download size: 340 k
Downloading Packages:
chrony-4.6.1-2.el9.x86_64.rpm                   3.5 MB/s | 340 kB     00:00
--------------------------------------------------------------------------------
Total                                           800 kB/s | 340 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: chrony-4.6.1-2.el9.x86_64                              1/2
  Upgrading        : chrony-4.6.1-2.el9.x86_64                              1/2
  Running scriptlet: chrony-4.6.1-2.el9.x86_64                              1/2
  Running scriptlet: chrony-4.5-1.el9.x86_64                                2/2
  Cleanup          : chrony-4.5-1.el9.x86_64                                2/2
```

client [Работает] - Oracle VirtualBox

Файл Машина Вид Ввод Устройства Справка

Activities Terminal Mar 12 08:39 en

vagrant@client:~

```
[vagrant@client ~]$ timedatectl
               Local time: Thu 2026-03-12 08:39:14 UTC
           Universal time: Thu 2026-03-12 08:39:14 UTC
                 RTC time: Thu 2026-03-12 08:39:14
                Time zone: UTC (UTC, +0000)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[vagrant@client ~]$ date
Thu Mar 12 08:39:18 AM UTC 2026
[vagrant@client ~]$ hwclock
hwclock: Cannot access the Hardware Clock via any known method.
hwclock: Use the --verbose option to see the details of our search for an access
 method.
[vagrant@client ~]$ hwclock
```

Rocky

Right Ctrl

Open Save

chrony.conf
/etc

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
pool 2.rocky.pool.ntp.org iburst

# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
 allow 192.168.0.0/16

# Serve time even if not synchronized to a time source.
#local stratum 10

# Require authentication (nts or key option) for all NTP sources.
#authselectmode require

# Specify file containing keys for NTP authentication.
keyfile /etc/chrony.keys

# Save NTS keys and cookies.
ntsdumpdir /var/lib/chrony
```

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```
[root@server ~]#
[root@server ~]#
[root@server ~]#
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]#
```

```
[vagrant@client ~]$ systemctl restart chronyd
Failed to restart chronyd.service: Connection timed out
See system logs and 'systemctl status chronyd.service' for details.
[vagrant@client ~]$ chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^? 51.250.35.68                  0   6     0     -     +0ns[   +0ns] +/-    0ns
^? 77.50.202.25                  0   6     0     -     +0ns[   +0ns] +/-    0ns
^? 91.188.214.68                 0   6     0     -     +0ns[   +0ns] +/-    0ns
^? 92.255.126.1                  0   6     0     -     +0ns[   +0ns] +/-    0ns
[vagrant@client ~]$
```

```
[root@server ~]#
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* 6.18.241.92.s-inform.net      3   6   377    50  -1080us[ -973us] +/-   12ms
^- 89.188.118.150                3   6   377    51  +1639us[+1747us] +/-  108ms
^+ time.cloudflare.com           3   6   377    51  +3099us[+3206us] +/-   16ms
^+ hcvvv7-real.sci-nnov.ru       2   6   377    51   -932us[ -825us] +/-   21ms
[root@server ~]#
```

client [Работает] - Oracle VirtualBox

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Activities Text Editor Mar 12 09:04 en

Open *ntp.sh /vagrant/provision/client Save

Could not save the file "/vagrant/provision/client/ntp.sh".

Unexpected error: Error renaming temporary file: Text file busy

```
#!/bin/bash

echo "Provisioning script $0"

echo "Copy configuration files"
cp -R /vagrant/provision/client/ntp/etc/* /etc
restorecon -vR /etc

echo "Restart chronyd service"
systemctl restart chronyd
```

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Right Ctrl

```
                run: "always",
                path: "provision/client/01-routing.sh"
    client.vm.provision "client ntp",
                type: "shell",
                preserve_order: true,
                path: "provision/client/ntp.sh"

    client.vm.provider :virtualbox do |v|
        v.linked_clone = true
# Customize the amount of memory on the VM
        v.memory = 4000
```

```
47      server.vm.provision "server ntp",
48                  type: "shell",
49                  preserve_order: true,
50                  path: "provision/server/ntp.sh"
51      server.vm.provider :virtualbox do |v|
52          v.linked_clone = true
53  # Customize the amount of memory on the VM
```

Вывод

В ходе выполнения лабораторной работы были получены навыки по управлению системным временем и настройке синхронизации времени.

Спасибо за внимание!